Nerve Segmentation Project (Total: 100 points)

Team Members:

1. **Dataset Description (5 points)**
    - Describe the dataset used, any issue.
    - Discuss any preprocessing steps (normalization, augmentation, resizing).
2. **Model Selection & Explanation (60 points)**
    - What model architecture did you use (e.g., U-Net, nnU-Net, DeepLab, custom CNN)?
    - Explain the structure of the model in detail. What is the main contribution of the model? What make this model work well?
    - If pretrained weights were used, discuss the source domain and how effective the knowledge transfer is.
3. **Experimental Design (25 points)**
    - What was your training strategy (loss function, optimizer, learning rate schedule)?
    - What metrics were used to evaluate performance?
    - Explain how you avoided overfitting (data augmentation, dropout, early stopping, cross-validation).
4. **Results & Analysis (10 points)**
    - Present quantitative results.
    - Provide visual examples of successful segmentations and cases where the model failed.
    - Compare baseline vs. improved experiments if applicable.